

SISEN Labs Student Guide

1 Purpose

This document provides the shared setup, prerequisites, and common working practices for the SISEN lab scenarios.

The SISEN labs examine how security failures in wireless, IoT, simulated BLE, MQTT, and low-power networked systems can become **safety-relevant system behaviours**. The focus is not only on observing an attack, but on understanding how a technical security event can affect system behaviour, operator interpretation, and safety outcomes.

Each scenario follows the same general pattern: students start the scenario, observe normal behaviour, capture traffic, run a controlled attack or failure demonstration, observe the system effect, and then map the observed behaviour to a possible hazard and mitigation strategy.

2 How to Use This Lab Set

- Start with this document to prepare the SISEN environment.
- Use the relevant scenario guide to start, observe, and stop the Medical IoT, Smart Building, or 6LoWPAN scenario.
- Use the attack, observation, and disruption guide for the controlled activities and evidence-capture commands.
- Use the hazard analysis worksheet to connect the observed security event to a possible safety consequence.
- You may also use the **README** files provided in the GitHub repository for additional setup notes and command examples.

3 Repository Access

- GitHub repository: `https://github.com/azelhajjar/SISEN.git`
- Recommended local path: `$HOME/Desktop/SISEN`

>_ Clone the repository

```
cd $HOME/Desktop
git clone https://github.com/azelhajjar/SISEN.git
cd SISEN
```

Repository Notes

The repository includes **README** files that explain the purpose of the main folders and scripts. Use these files alongside the lab handouts when you need additional command examples or folder-specific notes. All commands in the SISEN documentation assume that you are working from the root folder of the SISEN repository unless stated otherwise.

4 Lab Environment

SISEN should be run on a Debian-based Linux system. The system can be installed directly on a physical machine or run inside a virtual machine. The important requirement is the Linux environment, not whether the machine is native or virtualised.

The labs were tested on Ubuntu 26.04.

- Operating system: Debian-based Linux, such as Debian or Ubuntu.
- Installation type: native installation or virtual machine.
- Required privileges: some networking, namespace, AP, and capture commands require `sudo`.
- Network mode: NAT networking is sufficient when SISEN is run in a virtual environment.

4.1 Core Dependencies

SISEN uses a Debian-based Linux environment with Python, MQTT tooling, AP infrastructure tools, attack-support tools, and packet capture tools. The dependencies are grouped below so that each set has a clear purpose.

>_ Install required SISEN packages

```
sudo apt update
sudo apt install -y hostapd dnsmasq mosquitto mosquitto-clients iw iproute2 net-
tools python3 python3-venv python3-pip git curl dos2unix netcat-openbsd
```

>_ Install packet capture and analysis tools

```
sudo apt install -y tcpdump wireshark
```

4.2 Wi-Fi Infrastructure Attack Tools

Some SISEN lab activities involve controlled activities against the lab access point. These activities require Wi-Fi analysis and attack-support tools. Kali Linux usually includes these tools by default, but they may need to be installed manually on a standard Debian or Ubuntu system.

>_ Install WiFi infrastructure attack tools

```
sudo apt install -y aircrack-ng
```

Use these tools only against the controlled lab access point and only when instructed by the scenario document.

5 Initial Setup

After cloning the repository, **run the SISEN setup script**. The setup script prepares the Python environment, installs the required Python packages, checks the required tools, and handles any privileged commands internally when needed.

>_ Run the SISEN setup script

```
cd $HOME/Desktop/SISEN
./setup.sh
```

6 General Working Practice

- Record the scenario, AP mode, and lab activity being used.
- Observe the dashboard, terminal output, logs, and captures before and after the security event.

- Keep the lab running long enough to observe baseline behaviour before triggering the attack or failure demonstration.
- Continue observing the system after the attack to see whether the effect is temporary, persistent, or recoverable.
- Use your notes to complete the hazard analysis section of the scenario.

Common Lab Workflow

Each SISEN scenario follows the same workflow:

Stage	Student activity
Start scenario	Launch the assigned scenario and confirm that the required services are running.
Observe normal behaviour	Watch the dashboard, telemetry stream, logs, and network traffic before any security event is triggered.
Capture traffic	Start packet capture on the relevant interface or subscribe to the relevant MQTT topics.
Run controlled activity	Trigger the bounded observation, attack, or disruption activity described in the activity guide.
Observe system effect	Compare dashboard state, telemetry, logs, and traffic before and after the activity.
Map to hazard	Identify the safety-relevant consequence of the observed system behaviour.
Discuss mitigation	Propose controls, resilience mechanisms, and residual risk.

6.1 Capture-Friendly Execution

The labs are designed to be capture-friendly. In normal teaching mode, scenarios should remain running until stopped by the user.

Capture practice

Do not start and stop scenarios too quickly. You need time to observe normal behaviour, start packet capture, trigger the controlled activity, and compare the resulting evidence.

Typical evidence sources include:

- packet captures from wireless, namespace, low-power, or MQTT-facing interfaces;
- MQTT topic output from `mosquitto_sub`;
- dashboard state before, during, and after the security event;
- terminal logs from scenario runners, gateways, controllers, and attack helpers;
- hazard analysis notes completed during the scenario.

7 Security-Informed Safety Reasoning

The labs do not stop at identifying a security issue. You should reason from the security event to the system effect and then to the possible safety consequence.

Reasoning step	Example question
Security event	What attack, failure, spoofing, replay, suppression, or disruption occurred?
Affected component	Which sensor, gateway, broker, dashboard, AP, namespace, or trust boundary was affected?
Observed system effect	What changed in the dashboard, telemetry, logs, or captured traffic?
Safety-relevant hazard	What unsafe or misleading system behaviour could follow?
Mitigation	What control would reduce the likelihood or impact of this behaviour?
Residual risk	What risk remains even after the proposed mitigation?

8 Hazard Analysis Notes

You should complete a hazard analysis worksheet for each scenario. The worksheet should be completed during the lab, not only after the lab has finished. The hazard analysis worksheet is available [online](#) and in the SISEN repository under `teaching-materials/labs/`. The hazard analysis should capture:

- the scenario or lab name;
- the security event;
- the affected component or trust boundary;
- the observed system effect;
- the safety-relevant hazard;
- the evidence observed in the lab;
- existing controls;
- residual risk or discussion points.

Cleanup The main SISEN scenario scripts perform their normal cleanup when the scenario is stopped. Some activities, such as wireless monitor-mode capture, require an additional cleanup command. Follow the activity-specific stopping instructions where provided.